General
Finding Time Complexity o...
Problem 1: Finding Complexi...
Problem 2: Finding Complexi...
Problem 3: Finding Complexi...
Problem 4: Finding Complexi...
Problem 5: Finding Complexi...
Greedy Algorithms
1-G-Coin Problem
2-G-Cookies Problem
3-G-Burger Problem
4-G-Array Sum max problem
5-G-Product of Array elemen...
Divide and Conquer
1-Number of Zeros in a Give...
2-Majority Element
3-Finding Floor Value

Problem 1: Finding Complexity using Counter Method

Started on	Wednesday, 5 August 2026, 9:02 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:40 AM
Time taken	13 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

void function (int n)
{
    int i,s;
    int c=0;
     i =1;c++;
     s =1;c++;


    while(s <= n)
    {
        c++;

         i++;c++;
         s += i;c++;
     }
     c++;
     printf("%d",c);
}
int main(){
    int n;
    if(scanf("%d",&n)==1){
       function(n);
    }
   return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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Problem 2: Finding Complexity using Counter method

Started on	Wednesday, 12 August 2026, 8:35 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:40 AM
Time taken	7 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
      for(int i=1; i<=n; i++)
      {
        for(int j=1; j<=n; j++)
        {
           printf("*");
           printf("*");
           break;
        }
      }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, count = 0;
    scanf("%d", &n);

    if (n == 1)
    {
        printf("*");
    }
    else
    {
        count++;

        for (int i = 1; i <= n; i++)
        {
            count++;

            for (int j = 1; j <= n; j++)
            {
                count++;
                count++;
                count++;
                break;
            }

            count++;
        }

        count++;
    }

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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Problem 3: Finding Complexity using Counter Method

Started on	Wednesday, 12 August 2026, 8:24 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:40 AM
Time taken	7 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>

int main(){
   int num,i,c = 0;
   scanf("%d",&num);
   for(i=1;c++,i<=num;c++,++i){
     if(num%i==0){
       c++;
      }
   }
   printf("%d",c);
   return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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Problem 4: Finding Complexity using Counter Method

Started on	Wednesday, 12 August 2026, 8:44 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:41 AM
Time taken	6 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>

int main(){
    int c=0;
    int n;
    scanf("%d",&n);
    c++;
    for(int i=n/2;i<n;i++){
        c++;

        for(int j=1;j<n;j=2*j){
            c++;

            for(int k=1;k<n;k=k*2){
                c++;
                c++;
            }
            c++;

        }
        c++;

    }
    c++;

    printf("%d",c);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Problem 5: Finding Complexity using counter method

Started on	Wednesday, 19 August 2026, 8:11 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:41 AM
Time taken	29 mins 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int c =0;
void rev(int n){
    int re = 0,rem;
    c++;
    while(n!=0){
        rem = n%10;
        re = re*10+rem;
        n/=10;
        c+=3;
        c++;
    }
    c++;
}
int main(){
    int n;
    scanf("%d",&n);
    c++;
    rev(n);
    printf("%d",c);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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